

RINFORZO 16
10 esercizi simili in più

4. Equazioni numeriche fratte

GUARDA!
Le risorse di questa pagina

E

419 $\frac{3x-2}{x+4} = 5$

420 $\frac{12}{x-8} - \frac{6x}{8-x} = 0$

421 $\frac{x^2-2x+1}{3x-3} = 0$

422 $\frac{1}{x^2} - \frac{2}{3x} = \frac{7}{x}$

423 $\frac{4x-7}{x+3} = 2$

424 $\frac{5y}{y-1} = \frac{y+3}{y-1}$

425 $\frac{1+6x}{3x-4} = \frac{2x+3}{x}$

426 $\frac{10-a^2}{a+6} = 2-a$

427 $\frac{3x}{2x+9} - \frac{1}{2} = 0$

428 $\frac{8}{x-12} - 1 = 3$

429 $\frac{9}{x} + \frac{25}{2x} - \frac{1}{3x} = \frac{x+4}{2x}$

430 $\frac{9x-2}{3x-3} + \frac{x+2}{1-x} = 0$

431 $\frac{2x-5}{1-3x} = \frac{4x+15}{9x-3}$

432 $5x+100 = \frac{10000}{100-5x}$

433 $\frac{5x}{x-2} = \frac{x+4}{2} - \frac{x^2}{2x-4}$

434 $2 + \frac{1-x}{x^2} = \frac{-x+3(x+2)}{x}$

435 $\frac{3x}{4x-2} - \frac{2x}{6x-3} = \frac{5}{2x-1}$

436 $\frac{7x-5}{x^2} - \frac{x^2-2x+1}{3x} = \frac{-x+2}{3}$

437 $\frac{3x+2}{x} = \frac{4x}{3x-1} + \frac{5x^2-2}{3x^2-x}$

438 $\frac{3x+1}{x} = \frac{9x+3}{3x+1}$

439 $\frac{1+4x}{4x-4} + \frac{3}{2x} = \frac{x+1}{x-1}$

440 $\frac{9x^2+4}{6x^2-3x} = \frac{4x+1}{2x} - \frac{x+1}{2x-1}$

441 $\frac{x^2-3x+4}{3x^3+3x^2} - \frac{x^2-4x+8}{3+3x^3} = 0$

442 $\frac{9}{x^2+4x+4} = 1 - \frac{x}{x+2}$

443 $\frac{4x}{1+x} - 3 = \frac{3x-9}{3x+3}$

444 $1 - \frac{2x+4}{x-1} = \frac{2}{1-x}$

445 $3 - \frac{2}{x} + \frac{1}{x^2} = \frac{(x-1)(3x+1)}{x^2}$

446 $\frac{x-5}{x+3} - \frac{1}{2} = \frac{3}{4}$

447 $\frac{4-x^2}{3-x} = 1+x$

448 $\frac{2x}{9x-3} - \frac{3x}{6x-2} = \frac{7}{3x-1}$

449 $\frac{3}{5x} = \frac{1}{x-5} - \frac{4x+15}{10x^2}$

450 $\frac{2x+1}{3x} - \frac{1}{3x-x^2} = \frac{6x-5}{9x-27}$

451 $\frac{3}{x^2-2x} - \frac{1}{x} = -\frac{2}{2-x}$

452 $\frac{1}{2(x+1)} - \frac{2}{(1+x)(1-x)} = \frac{x}{2(1-x)^2}$

453 $\frac{3(x+5)}{x^2-9} - \frac{2x+5}{x+3} = \frac{2x-4}{3-x}$

454 $\frac{(x-3)^2}{x-2} + \frac{15}{3} = \frac{x^2-x-1}{x-2}$

455 $\frac{x+2}{x-1} - \frac{2x}{x^2-1} = 1 + \frac{3}{x-1}$

456 $\frac{x(x-3)+2}{x-1} = x-2$

457 $\frac{1}{2x^2} - \frac{x+3}{4x^2-1} = \frac{1+x}{3x-6x^2} - \frac{1}{12x}$

458 $\frac{4-3x}{3x} = \frac{11-3x}{3x+12} - \frac{3(7-19x)}{2x^2+8x}$

459 $\frac{x+3(1-2x)}{x^2+4x+4} - \frac{3x}{x+2} = -3$

460 $\frac{1}{3x+3} - \frac{2}{x-7} = \frac{16}{6x+7-x^2}$

461 $\frac{x+16}{x^2-3x-10} + 2 = \frac{x-2}{x-5} + \frac{x}{x+2}$

462 $\frac{13-6x}{2x+5} - \frac{8-3x}{x-1} = \frac{15x-1}{4x^2+6x-10}$

463 $\frac{1}{x-1} + \frac{x+4}{x^2+2x-3} = \frac{25+11x+2x^2}{(x-1)(x+3)^2}$

464 $\left(\frac{2-x}{2+x} - 1\right)\left(\frac{x+1}{2+x} - 1\right) = \frac{4}{x^2+4x+4}$

465 $\frac{1}{x^2+2x} - \frac{2}{x^2+x-2} = -\frac{1}{x^2-2x+1}$

466 $\frac{9x-4}{x-2} = \left(3 + \frac{2}{x-2}\right)\left(3 - \frac{2}{x-2}\right)$

467 $\frac{1-x^2}{2x^3-16} = \frac{4x-3}{x^2+2x+4} + \frac{9}{4-2x}$

468 $\frac{x^2+x-2}{4x-4} = \frac{x^3+1}{4+4x^2}$

469 $\frac{5}{x^2-3x} + \frac{2}{x^2-9} = \frac{5}{3-x} + 1 - \frac{x^3-5x^2+1}{x^3-9x}$

470 $-\frac{8x+7}{2x+1} + \frac{2(4+7x+2x^2)}{(x-2)(1+2x)} + 2 = 0$

435 $\frac{3x}{4x-2} - \frac{2x}{6x-3} = \frac{5}{2x-1}$

C.E

$4x-2 \neq 0 \quad (2x-1) \neq 0 \quad \boxed{x \neq \frac{1}{2}}$

$6x-3 \neq 0 \quad x = \frac{3}{6} = \frac{1}{2}$

$2x-1 \neq 0 \quad x = \frac{1}{2}$

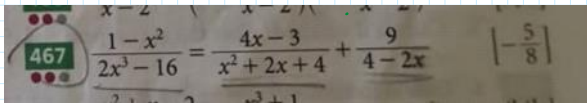
$$\frac{3x}{2(2x-1)} - \frac{2x}{3(2x-1)} = \frac{5}{2x-1}$$

$$\left. \begin{array}{l} 4x-2 = 2(2x-1) \\ 6x-3 = 3(2x-1) \\ (2x-1) \end{array} \right\} \begin{array}{l} \text{m.c.m.} \\ (2x-1) \cdot 2 \cdot 3 \end{array}$$

$$\frac{9x-4x}{6(2x-1)} = \frac{30}{6(2x-1)}$$

$$5x = 30$$

$$x = 6$$



$$\frac{1-x^2}{2(x-2)(x^2+2x+4)} - \frac{4x-3}{x^2+2x+4} + \frac{9}{2(x-2)} = 0$$

$$\frac{1-x^2 - [(4x-3)(2(x-2))] + 9(x^2+2x+4)}{2(x-2)(x^2+2x+4)} = 0$$

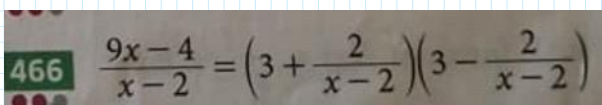
$$1-x^2 - [(4x-3)(2x-4)] + 9x^2 + 18x + 36 = 0$$

$$+8x^2 - 16x - 6x + 12$$

$$1 - x^2 - 8x^2 + 22x - 12 + 9x^2 + 18x + 36 = 0$$

$$40x + 25 = 0$$

$$x = \frac{-25}{40} = -\frac{5}{8}$$



$$\frac{9x-4}{x-2} = 9 - \frac{4}{(x-2)}$$

C.E

$$2x^3-16 \neq 0 \quad 2(x^3-8) \neq 0 \quad 2(x-2)(x^2+2x+4) \neq 0$$

$$x^2+2x+4 \neq 0 \quad (x^2+2x+1)+3 \neq 0$$

$$(x+1)^2 + 3 \neq 0 \rightarrow \text{somme} > 0$$

$$4-2x \neq 0 \quad -2(x-2)$$

$$x-2 \neq 0$$

$$x = 2$$

C.E.

$$x-2 \neq 0 \quad x \neq 2$$

$$\frac{9x-4}{x-2} = 9 - \frac{4}{(x-2)}$$

$$\frac{(9x-4)(x-2)}{(x-2)^2} = \frac{9(x-2)^2 - 4}{(x-2)^2}$$

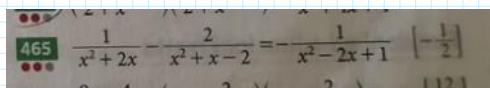
$$(9x-4)(x-2) = 9(x-2)^2 - 4$$

$$9x^2 - 18x - 4x + 8 = 9(x^2 - 4x + 4) - 4$$

$$9x^2 - 22x + 8 = 9x^2 - 36x + 36 - 4$$

$$+14x = +28$$

$$x = \frac{28}{14} = \frac{12}{7}$$



C.E.

$$x^2 + 2x \neq 0 \quad x(x+2) \neq 0 \quad \begin{matrix} x \neq 0 \\ x+2 \neq 0 \end{matrix} \quad \begin{matrix} x \neq -2 \end{matrix}$$

$$x^2 + x - 2 \neq 0 \quad (x-1)(x+2) \neq 0 \quad \begin{matrix} x-1 \neq 0 \\ x+2 \neq 0 \end{matrix} \quad \begin{matrix} x \neq 1 \end{matrix}$$

$$x^2 - 2x + 1 \neq 0 \quad (x-1)^2 \neq 0 \quad \begin{matrix} x-1 \neq 0 \\ x \neq 1 \end{matrix}$$

$$\frac{1}{x(x+2)} - \frac{2}{(x-1)(x+2)} + \frac{1}{(x-1)^2} = 0$$

$$\frac{(x-1)^2 - 2[x(x-1)] + x(x+2)}{x(x+2)(x-1)^2} = 0$$

$$(x-1)^2 - 2[x(x-1)] + x(x+2) = 0$$

$$x^2 - 2x + 1 - 2x^2 + 2x + x^2 + 2x = 0$$

$$2x + 1 = 0$$

$$x = -\frac{1}{2}$$